



SIR OTONDE MICHAEL AWITI
KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
SMA 112 : BASIC MATHEMATICS

1ST SEMESTER 2023/2024

TIME: 2 HRS

INSTRUCTIONS

Answer Question ONE and any other Two Questions

SIR OTONDE
MICHAEL AWITI

QUESTION 1 (30 MARKS)

(a) When $x^3 + 2x^2 + ax + 7$ is divided by $x - 2$ the remainder is 3. Find a . (2 Marks)

(b) Find the set of values of x for which $15 + x - 2x^2 > 0$. (4 Marks)

(c) How many even numbers of four digits can be formed by using the figures 1, 2, 4, 8 if repetitions are allowed? (2 Marks)

SIR OTONDE MICHAEL AWITI

(d) Out of 100 lorries stopped by police at a road block, 60 were passed as satisfactory.

Each of the remaining 40 had faults in brakes (B), Lights (L) and Steering (S). The police found that

$$n(B) = 12, n(B \cap L \cap S) = 3, n(B^c \cap L \cap S) = 2, n(L \cap S) = n(B \cap S),$$

$$n(B \cap L \cap S^c) = 1, n(B^c \cap L^c \cap S) = n(B^c \cap L \cap S^c).$$

How many lorries had only one fault and how many had faulty lights? (5 Marks)

(e) Use Boolean algebra to simplify $\sim(p \Rightarrow q) \vee p$. (5 Marks)

(f) Prove by mathematical induction that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

SIR OTONDE MICHAEL AWITI (6 Marks)

(g) Express $\frac{6+3i}{10+8i}$ in the form $a + bi$, where a, b are real numbers. (3 Marks)

(h) Find the eighth term in the expansion of $\left(2x^2 - \frac{1}{x^2}\right)^{12}$. (3 Marks)

SIR OTONDE MICHAEL AWITI

79/16

SIR OTONDE MICHAEL AWITI

QUESTION 2 (15 MARKS) X

(a) Let $A = \{n \in \mathbb{N} | n \leq 11\}$, $B = \{n \in \mathbb{N} | n \text{ is even and } n \leq 20\}$ and

$E = \{n \in \mathbb{N} | n \text{ is even}\}$. Find

(i) $A - B$

SIR OTONDE MICHAEL AWITI (1 Mark)

(ii) $A \Delta E$

(1 Mark)

(iii) $\mathbb{N} - E$

(1 Mark)

(b) Use Boolean algebra to simplify $(A \cap B) \cup (A^c \cap B) \cup (A \cap B^c) \cup (A^c \cap B^c)$

(4 Marks)

(c) Given the sets $A = \{1, 2, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{2, 4, 6, 8\}$ and the universal

set $U = \{1, 2, \dots, 9\}$, find $A^c \cap (B \cup C)$.

(2 Marks)

(d) Use the truth table for $(p \Rightarrow q) \Leftrightarrow (\sim p \vee q)$ and decide whether the statement is a

tautology, a contradiction or neither.

(3 Marks)

(e) State the converse, inverse and the contrapositive of the statement 'If a triangle is

equiangular, then it is equilateral'.

(3 Marks)

SIR OTONDE MICHAEL AWITI

QUESTION 3 (15 Marks)

(a) If α and β are roots of the equation $5x^2 - x - 2 = 0$, find without solving the

equation the quadratic equation whose roots are $1 - \frac{\alpha}{\beta}$ and $1 - \frac{\beta}{\alpha}$. (4 Marks)

(b) Factorize $f(x) = 2x^3 + x^2 - 8x - 4$. Hence solve the equation $f(x) = 0$.

SIR OTONDE MICHAEL AWITI

(5 Marks)

(c) A polynomial $f(x)$ has remainder 4 when divided by $x - 2$ and remainder 7 when

divided by $x - 3$. Find the remainder when $f(x)$ is divided by $(x - 2)(x - 3)$.

(3 Marks)

(d) Find the set of values of x for which $6 + x > 12x^2$.

(3 Marks)

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

QUESTION 4

(15 MARKS) ✓

- (a) In how many ways can letters of the word BESIEGE be arranged (2 Marks)
- (b) A mixed hockey team containing 5 men and 6 women is to be chosen from 7 men and 9 women. In how many ways can this be done. (3 Marks)
- (c) Prove by induction that $1^3 + 2^3 + \dots + n^3 = \frac{1}{4} n^2(n+1)^2$ (5 Marks)
- (d) Find the coefficient of x^{10} in the expansion of $(2x - 3)^{14}$ (3 Marks)
- (e) Prove that for all real numbers x , $x^2 - 4x + 17 \neq 0$ (2 Marks)

QUESTION 5

(15 MARKS) ✓

- (a) Find the four roots of $z^4 = 8(-1 + i\sqrt{3})$. Leave your answer in the form $a + bi$, $a, b \in \mathbb{R}$. (5 Marks)
- (b) Calculate $\left\{2\left(\cos \pi/5 + i \sin \pi/5\right)\right\}^5$. (4 Marks)
- (c) Express $z = 1 + i$ in polar form. (3 Marks)
- (d) Write the following complex number $\frac{2-9i}{5-2i}$ in the form $x + iy$. (3 Marks)

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$